



THE PROBLEM(S) WITH WEBASSEMBLY

And why I'm still optimistic
about it



Hi, I'm Divya

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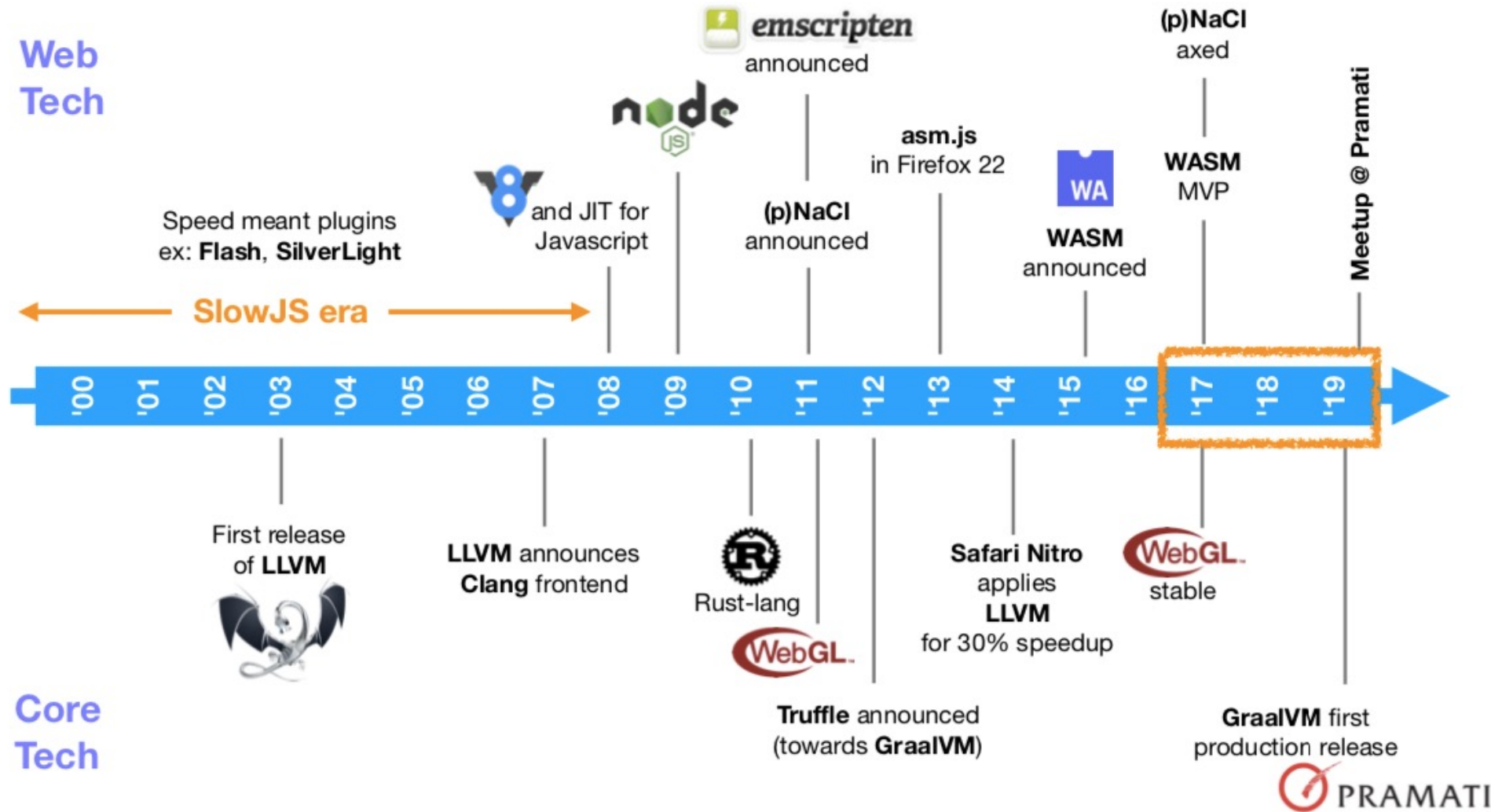
CNCF Ambassador



The image features a dark blue background with a subtle gradient. In the center, the text "Why WebAssembly?" is written in a white, handwritten-style font. Surrounding the text are six teal-colored brushstroke shapes, arranged in two columns of three. The shapes are irregular and have a hand-drawn appearance, with some overlapping the text area. The overall aesthetic is clean and modern, typical of a presentation slide.

Why WebAssembly?

Web Tech



Core Tech



chrislynch.link
<https://www.chrislynch.link> › tech › why-i-think-weba...

Why I Think WebAssembly is a Bad Idea - Chris Lynch

Yes, WebAssembly is good at "heavy lifting". Here's a great example of a project made better with WebAssembly. It's a clever piece of software that helps ...



bitsrc.io
<https://blog.bitsrc.io> › ...

What's Wrong with Web Assembly?. Does WASM ...

05-Jan-2021 — I think the answer to that question is: **poor marketing**. If you think about it, the web is littered with amazing projects that never made it ...

▲ c0nfused on Dec 17, 2015 | parent | context | favorite | on: [Compiling to WebAssembly: It's Happening](#)
Web assembly always makes me a little sad. It feels like we are going back to flash only it won't be bad this time, I promise, no really.
I always feel like the most obvious use for it is to start writing truly hateful and abusive code.
I'm sure this is because I'm getting old.



Better Programming
<https://betterprogramming.pub> › its-2023-and-webasse...

It's 2023 and WebAssembly is Still Not Ready. Or is It?

23-Jan-2023 — During the last few years, however, a lot of things happened. Some of those things were **good** for web development: numerous improvements to the ...

What is Wasm?

```
mirror_mod = modifier_ob.  
set mirror object to mirror.  
mirror_mod.mirror_object =  
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False  
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True  
  
selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier_ob.  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select  
  
print("please select exactly  
  
-- OPERATOR CLASSES ----  
  
types.Operator):  
X mirror to the selected  
object.mirror_mirror_x"  
mirror X"  
  
context):  
context.active_object is not
```

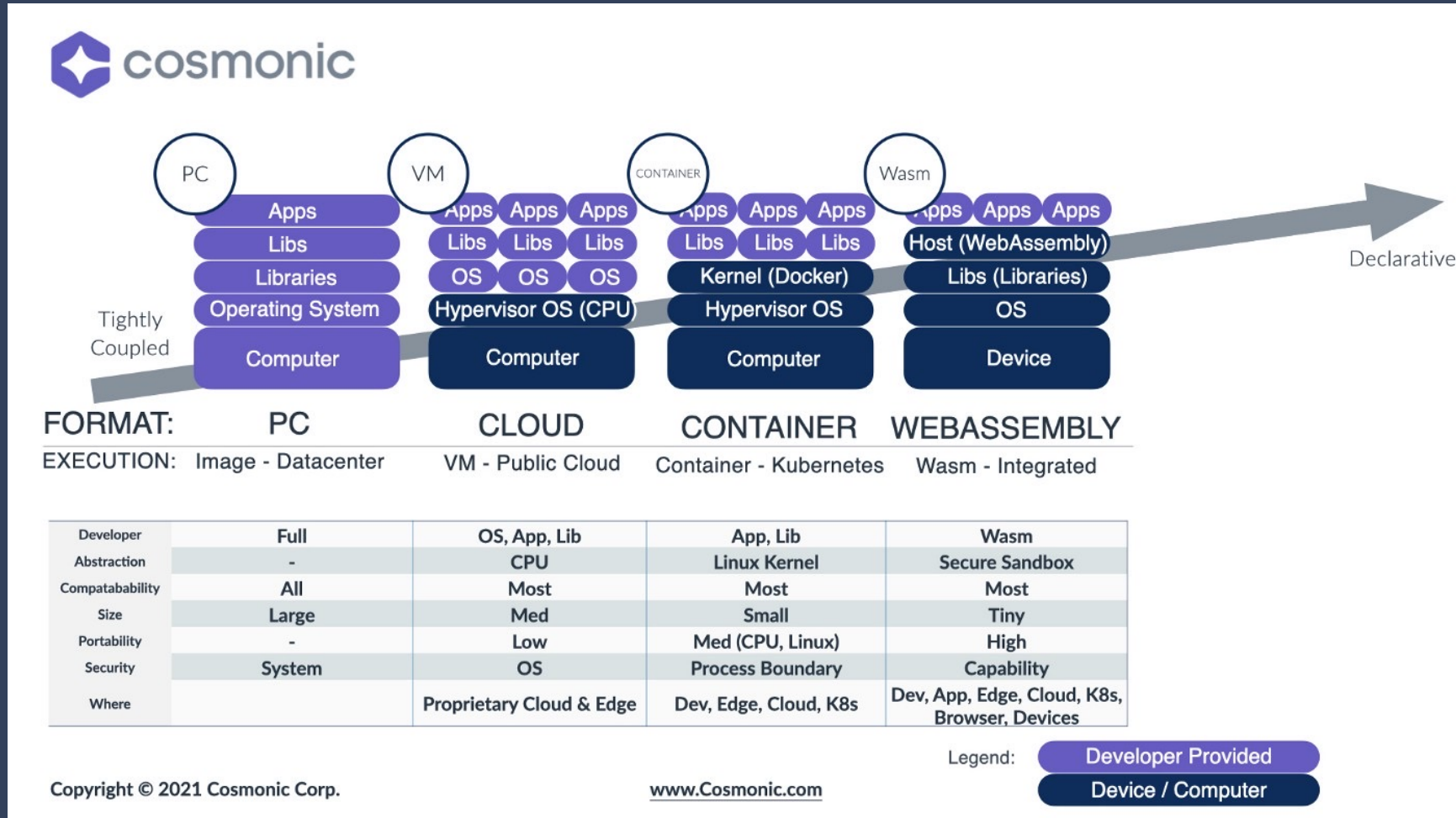
“WebAssembly (abbreviated *Wasm*) is a binary instruction format for a stack-based virtual machine. Wasm is designed as a portable compilation target for programming languages, enabling deployment on the web for client and server applications.”

-- <https://webassembly.org>

Q1. Why are we reinventing
the wheel?



Server side story



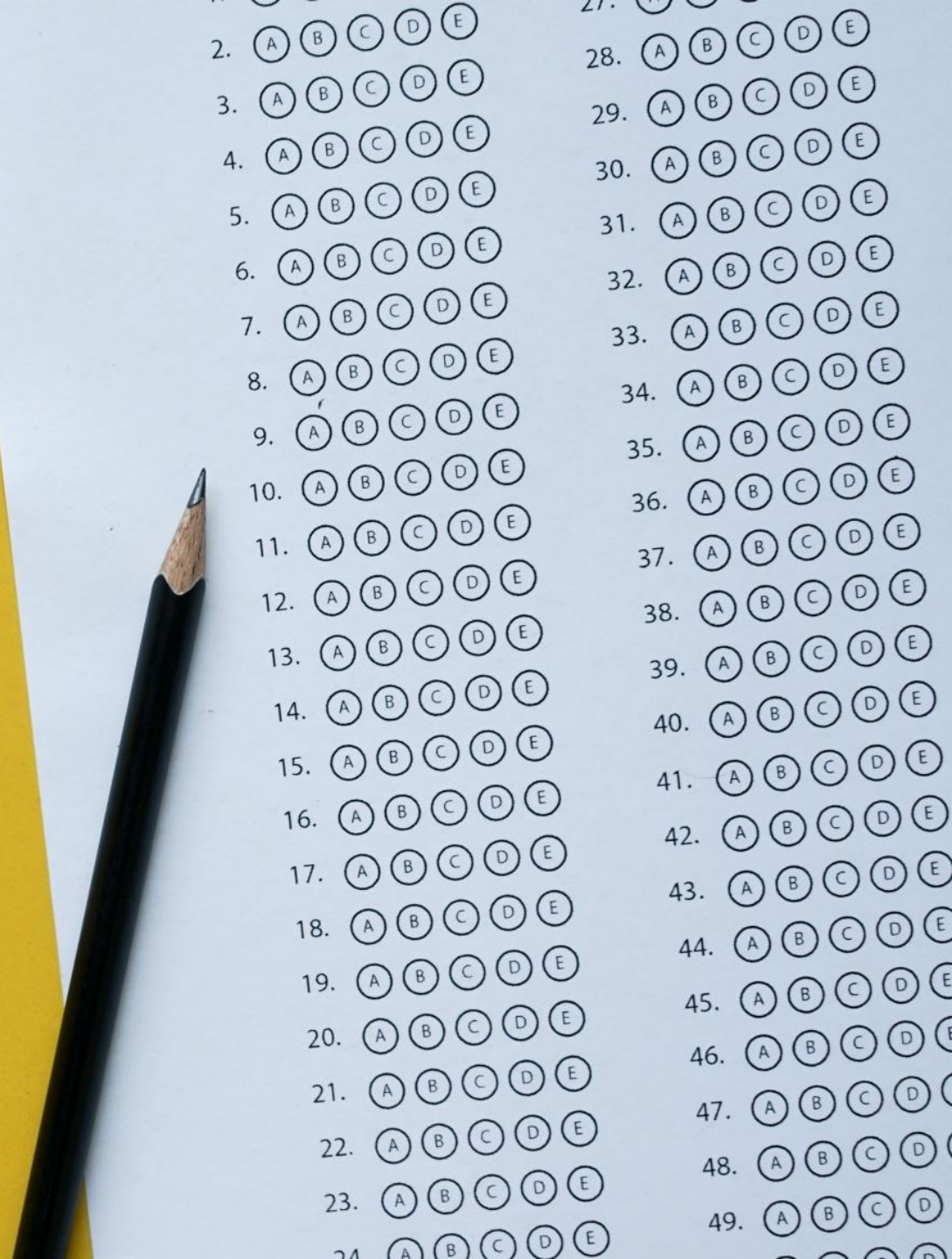
Source: <https://cosmonic.com/blog/kubecon-eu-2021-webassembly-keynote-cloud-native-and-wasm-better-together/>

The front end story





Q2. What about the DevEx?



Great Developer Experience



Onboarding curve



Tooling ecosystem



Awareness & Education



ONBOARDING

Language	Core	Browser	WASI	Spin SDK
JavaScript	✓	✓	⚡	✓
Python	✓	⚡	✓	⚡
Java	✓	✓	✓	⚡
PHP	✓	✓	✓	✗
CSS	N/A	N/A	N/A	N/A
C# and .NET	✓	✓	✓	✓
C++	✓	✓	✓	✗
TypeScript	✓	⚡	✗	✓
Ruby	✓	✓	✓	✗
C	✓	✓	✓	✗
Swift	✓	✓	✓	✗
R	✗	✓	✗	✗
Objective-C	?	✗	✗	✗
Shell	N/A	N/A	N/A	N/A
Scala (JVM)	✓	✓	✓	⚡
Scala (native)	⚡	✗	✗	✗
Go	✓	✓	✓	✓
PowerShell	✗	✗	✗	✗
Kotlin (JVM)	✓	✓	✓	⚡
Kotlin (Native)	⚡	✓	⚡	✗
Rust	✓	✓	✓	✓

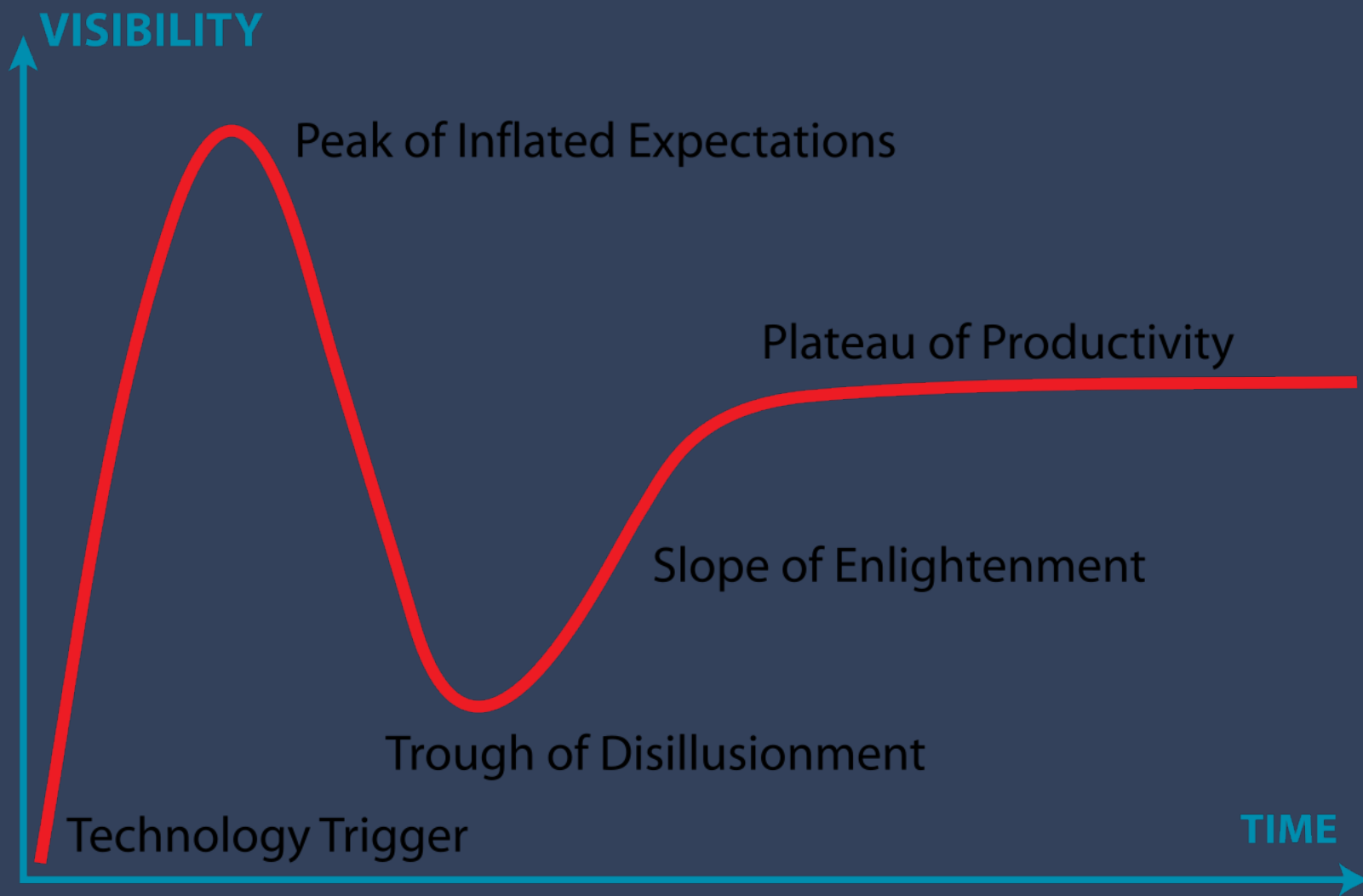
Source: <https://www.fermyon.com/wasm-languages/webassembly-language-support>

Tooling ecosystem



DO YOU HAVE TIME TO TALK ABOUT

WEBASSEMBLY





CLOUD NATIVE
Wasm DAY
EUROPE

Welcome + Opening Remarks

Kelsey Hightower, Google

<https://youtu.be/wZQikdoAG1I>



The Component Model



Thank you!